MNL Market Share Models

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A key idea in modern thinking about competition is that a buyer picks the alternative which appears to offer the greatest buyer's surplus. The surplus b a buyer gains is $b = w - p$, where w is the buyers willingness-to-pay (WTP) for the product or service and p is the price. The profit a seller gains through such a sale is $\pi = p - c$, where c is the seller's cost. This way of looking at the situation neatly combines the concepts of price, cost, and quality.

Consider a simple situation in which three products, labeled (1, 2, and 3) compete with one another. The prices and buyers' WTPs are shown in Exhibit 1. Faced with this situation, a buyer would choose product 1. If every buyer did so, product 1 would capture 100% of the market and no one would buy products 2 or 3.

Exhibit 1—Three Competing Products

	Product 1	Product 2	Product 3
WTP (w)	25	24	26
Price (p)	10	10	14
Buyer Surplus (b)	15	14	12

We know this is an unrealistic conclusion. In real markets, small differences in value or price do not produce such extreme outcomes. If we keep the assumption that buyers choose the alternative offering the greatest perceived surplus, then it must be that buyers differ from one another in how they see the situation. That is, buyers choosing product 2 perceive the surplus attached to product 2 as being larger than that attached to product 1. Such differences in perceived surplus could be due to individual differences in WTP, differences in retail prices, differences in perceived net life-time prices, or errors in perception.

With this insight, the surplus perceived by buyer j associated with product k can be written as

$$b_{kj} = w_k - p_k + \varepsilon_{kj} = b_k + \varepsilon_{kj} \tag{1}$$

where $b_k = w_k - p_k$, is the pre-deviation surplus and ε_{kj} is the *deviation* in perceived surplus. These deviations will play a role in determining which buyer buys which product. For example, products 1 and 2 have bs of 15 and 15, giving product 1 a $1 advantage. But if buyer j experiences deviations $\varepsilon_{1j} = 0,\ \varepsilon_{2j} = +2$, then product 2 will be chosen.

Obviously, there are an uncountable number of ways to define or model such deviations. They could depend upon things like income, gender, location, age; they could flow from categories of product (e.g., compact versus full-size); they could be "random" with distribution shapes depending

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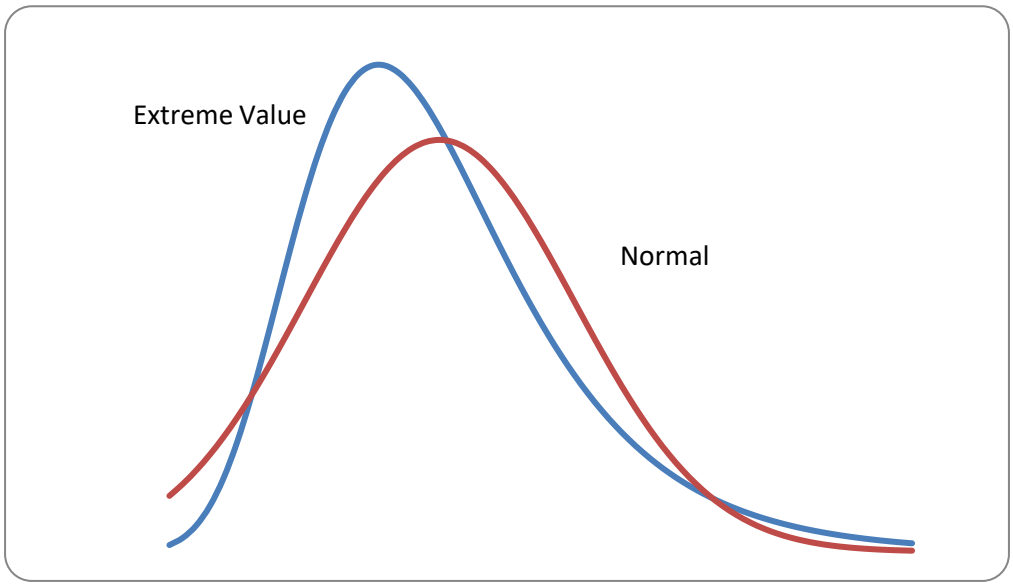
Exhibit 2—Normal and Extreme Value Distributions (m = 0, σ=1)



upon buyer or product characteristics. The very simplest possible assumption is that the deviations do not depend on the buyer or product—that realizations of ε_{jk} are identically and independently distributed.

The next hurdle is making an assumption about the shape of the distribution of the deviations. The usual assumption would be that disturbances, or "errors," they are Normal. However, using the Normal distribution to model these deviations leads to intractable models and difficult computations. However, if it is assumed that the deviations are distributed according to an extreme-value distribution, then a simple model of market shares is produced—the multinomial logit (MNL) model. Exhibit 2 displays both the Normal and the extreme value densities. You can see that the extreme value density is skewed and has a fatter right tail. For reference purposes, the extreme-value distribution with mean zero has the form

$$\Pr(\varepsilon \le X) = F(X) = e^{-e^{-(X/\mu+\gamma)}} \tag{2}$$

where μ is the measure of dispersion and the symbol γ is Euler's constant and is equal to 0.57722. The variance is given by

$$\sigma_\varepsilon^2 = \mu^2\pi^2 / 6. \tag{3}$$

We use this rather cumbersome distribution because it produces a neat clean result. The market share of product *k* (the probability that a buyer will choose *k*) is

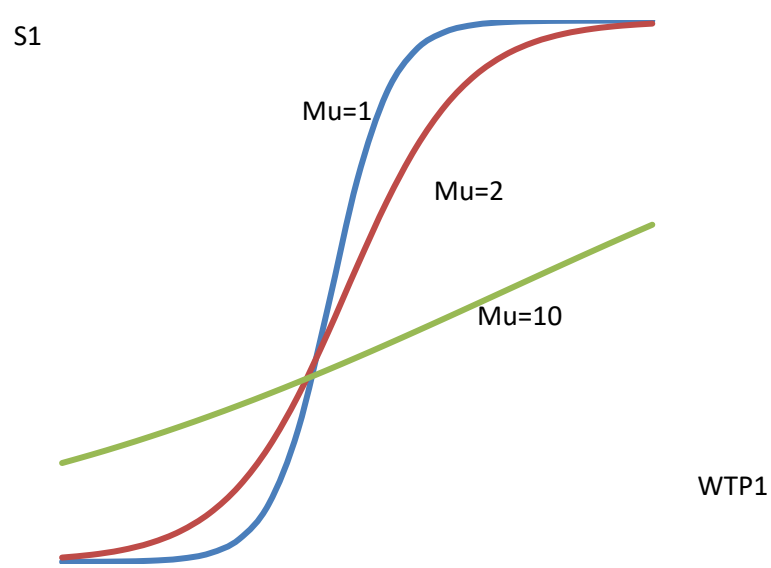
$$S_k = \frac{e^{b_k/\mu}}{\sum_{i=1}^{n} e^{b_i/\mu}} \tag{4}$$

Exhibit 3—Three Competing Products

Product	**1**	**2**	**3**
WTP (w)	25	24	26
Price (p)	10	10	14
Buyer Surplus (b)	15	14	12
Shares(μ =1)	70.5%	25.9%	3.5%
Shares(μ =2)	54.7%	33.1%	12.2%
Shares(μ =10)	37.8%	34.2%	28.0%

Exhibit 3 shows predicted market shares among our three competing products for μ = 1, 2, and 10. As can be seen, larger values of μ (larger σ_ε^2) pushes outcomes towards equality because the large disturbances swamp the "true" product differences in price and value. On the other hand, a small value of μ (small σ_ε^2) magnifies the impact of the differences among the b_k. As $\mu \to 0$, the product with the largest b_k captures the whole market. Exhibit 4 shows the relationship between product 1's WTP and its share is a logistic curve and how that curve changes with shifts in μ.

Exhibit 4—Share vs. Surplus Logistic Functions



Define the attractiveness of a product as

$$A_k = e^{b_k/\mu}. \tag{5}$$

The market share of product k is its share of total attractiveness:

$$S_k = \frac{A_k}{\sum_{i=1}^{n} A_i} \tag{6}$$

Note that attractiveness is purely a relative measure. Multiply every A_k by the same number and the market shares will be unchanged. As a consequence, the ratio of any two market shares is the ratio of the attractiveness measures:

$$\frac{S_j}{S_m} = \frac{A_j}{A_m}. \tag{7}$$

Take the natural logarithm of (7) yields

$$\begin{aligned} \ln\left[\frac{S_j}{S_m}\right] &= (b_j - b_m)/\mu \\ &= \frac{1}{\mu}\left[(w_k - w_m) - (p_k - p_m)\right] \end{aligned} \tag{8}$$

telling us that share ratios are determined by differences in buyer's surplus between offerings, scaled by μ.

Fitting an MNL Model

There is a vast literature and many software products aimed at "estimating" multinomial-logit models. What most of this treats are methods of using fine-grained data to study how various product and buyer characteristics influence the probability of purchase. This is the domain of the marketing engineer and this note will not venture into that territory. Rather, our interest here is in using rough information or even judgments about a market to build a multinomial logit model. That model can then be used to study competition and possibly answer certain kinds of "what if?" questions.

Suppose we have share and price data about three products as shown in Exhibit 5. Worksheet MNL3Products.xlsx contains these numbers and the following calculations.

We would like to fit an MNL model to this data by determining the three willingness-to-pay numbers (w_k) and μ. However, from (8) we see that only differences between levels of surplus matter (scaled by μ). This means that the MNL model does not care about the absolute levels of willingness-to-pay, only about the differences between alternatives. As a practical matter, we can either work just with differences or arbitrarily assign a value to one of the w_k.

Exhibit 5—Three Products

Product	1	2	3
Input Data			
Share	0.5	0.4	0.1
Price	25	32	20

Fixing one of the three w_k leaves 3 parameters yet to be determined—the two remaining w_k and μ. The standard approach in the marketing literature is to treat μ as a "scale" parameter and set it equal to 1. We will start with this assumption and see where it leads.

The best way to fit these parameters is to use (8):

$$(w_1 - 0) - (25 - 20)/1 = 1.609$$
$$w_1 = 6.6$$
$$(w_2 - 0) - (32 - 20)/1 = 1.386$$
$$w_2 = 13.386$$

That is, the MNL model fitting this data implies that WTP for product 1 is \$6.61 larger than the WTP for product 3. And, that the WTP for product 2 is \$13.39 greater,

This "solution" delivers back the three market shares. But, look at the demand elasticity faced by product 1. A little calculus will show that the MNL model implies

$$\begin{aligned} \frac{\partial S_k}{\partial p_k} &= -\frac{\partial S_k}{\partial b_k} = -S_k(1 - S_k)/\mu \\ E_{S_k, p_k} &= -\frac{p_k}{S_k}\frac{\partial S_k}{\partial b_k} = \frac{p_k}{\mu}(1 - S_k) \end{aligned} \tag{9}$$

Where E_{S_k, p_k} is the elasticity of share (or demand in units) of product *k* with respect to its price. As you can see, it depends on the ratio of price to μ, so setting μ=1 is hardly innocuous. Here, it implies that E_{S_k, p_k} = 25(.5) = 12.5, a very large value.

What, exactly, is μ, and how should it be set? This parameter is a bit confusing because it combines both scale and variance. To get some intuition, notice that (9) means that a product's price elasticity falls with share, reaching zero when share become 1.0 For a product with a 50% market share, if it takes a Δb increase in buyer's surplus to push up demand by 1%, then $\Delta b = 50\mu$. (Note that a 1% increase in demand is a ½ point increase in share.)

Returning to our problem, suppose that the analyst judges that product 1's own-price elasticity is probably about 1.2. That would imply $\mu = (1 - .5)(25)/1.2 = 10.4$. This information changes our computations of the WTP differences:

$$w_1 - w_3 = 5 + 1.609 \cdot 10.4 = 21.74$$
$$w_2 - w_3 = 12 + 1.386 \cdot 10.4 = 26.42$$

Armed with these WTP differences, we can perform a "What If" analysis. If the WTP of product 1 could be increased by $2, what would happen? Straightforward computations (See MNL3Products.xlsx) have product 1 increasing its share from 50 to 55%. If we had not adjusted μ, leaving it at 1.0, the larger implicit elasticity would have predicted a much sharper increase.